Data Analyst Project

By Sankalp Singh

OLA Data Analyst Project

Project Title: OLA Data Analyst Project

By: Sankalp Singh

Project Summary:

The **OLA Data Analyst Project** was designed to analyze and visualize booking data for OLA, one of India's leading ride-hailing platforms. The goal was to derive insights on ride performance, customer behavior, vehicle utilization, payment trends, and service quality using SQL and Power BI.
The project was divided into two core sections:

1. **Data Extraction and Analysis Using SQL**
2. **Data Visualization and Dashboarding Using Power BI**

Key Work Done:

1. SQL-Based Data Analysis:

- **Data Preparation:** Created views to extract key insights from the bookings dataset.
- **Ride & Customer Insights:**
  - Retrieved all successful bookings.
  - Identified top 5 customers based on ride count.
  - Calculated total successful booking value.
- **Ride Quality Metrics:**
  - Analyzed average ride distance by vehicle type.
  - Compared average customer and driver ratings across vehicle types.
  - Identified max and min driver ratings for Prime Sedan.
- **Cancellation Analysis:**
  - Counted cancellations by customers and drivers.
  - Extracted reasons for incomplete rides.
- **Payment & Revenue Insights:**
  - Filtered rides paid via UPI.
  - Aggregated booking values.

2. Power BI Dashboard Creation:

- Designed an interactive dashboard with the following visuals:
  - **Time-based Trends:** Ride volume over time.
  - **Status Breakdown:** Booking status proportions.
  - **Vehicle Analysis:** Ride distance by vehicle type, average ratings.
  - **Cancellation Reasons:** For both drivers and customers.
  - **Revenue Visualization:** By payment methods and top-paying customers.
  - **Rating Distributions:** Including customer vs. driver comparison.

Skills Demonstrated:

- **Data Analysis**: SQL querying, data aggregation, view creation.
- **Data Visualization**: Interactive dashboards in Power BI.
- **Critical Thinking**: Identifying KPIs, deriving actionable insights.
- **Communication**: Translating data into visual stories.

Tools & Technologies Used:

- **SQL (MySQL)** – for data manipulation and analysis.
- **Power BI** – for building comprehensive dashboards and reports.
    - **MS Excel** – for data organization (assumed based on standard practice).

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SQL Questions:

1. Retrieve all successful bookings:
2. Find the average ride distance for each vehicle type:
3. Get the total number of cancelled rides by customers:
4. List the top 5 customers who booked the highest number of rides:
5. Get the number of rides cancelled by drivers due to personal and car-related issues:
6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
7. Retrieve all rides where payment was made using UPI:
8. Find the average customer rating per vehicle type:
9. Calculate the total booking value of rides completed successfully:
10. List all incomplete rides along with the reason:

Power BI Questions:

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

Data Columns

1. Date
2. Time
3. Booking_ID
4. Booking_Status
5. Customer_ID
6. Vehicle_Type
7. Pickup_Location
8. Drop_Location
9. V_TAT
10. C_TAT
11. cancelled_Rides_by_Customer
12. cancelled_Rides_by_Driver
13. Incomplete_Rides
14. Incomplete_Rides_Reason
15. Booking_Value
16. Payment_Method
17. Ride_Distance
18. Driver_Ratings
19. Customer_Rating

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Power BI Answers:

Segregation of the views:

1. **Overall**
- Ride Volume Over Time
- Booking Status Breakdown

2. **Vehicle Type**
- Top 5 Vehicle Types by Ride Distance

3. **Revenue**
- Revenue by Payment Method
- Top 5 Customers by Total Booking Value
- Ride Distance Distribution Per Day

4. **Cancellation**
- Cancelled Rides Reasons (Customer)
- cancelled Rides Reasons(Drivers)

5. **Ratings**
- Driver Ratings
- Customer Ratings

Answers:

1. **Ride Volume Over Time:** A time-series chart showing the number of rides per day/week.
2. **Booking Status Breakdown:** A pie or doughnut chart displaying the proportion of different booking statuses (success, cancelled by the customer, cancelled by the driver, etc.).
3. **Top 5 Vehicle Types by Ride Distance:** A bar chart ranking vehicle types based on the total distance covered.
4. **Average Customer Ratings by Vehicle Type:** A column chart showing the average customer ratings for different vehicle types.
5. **cancelled Rides Reasons:** A bar chart that highlights the common reasons for ride cancellations by customers and drivers.
6. **Revenue by Payment Method:** A stacked bar chart displaying total revenue based on payment methods (Cash, UPI, Credit Card, etc.).
7. **Top 5 Customers by Total Booking Value:** A leaderboard visual listing customers who have spent the most on bookings.
8. **Ride Distance Distribution Per Day:** A histogram or scatter plot showing the distribution of ride distances for different Dates.
9. **Driver Rating Distribution:** A box plot visualizing the spread of driver ratings for different vehicle types.
10. **Customer vs. Driver Ratings:** A scatter plot comparing customer and driver ratings for each completed ride, analyzing correlations.

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SQL Questions & Answers

Create Database Ola;
Use Ola;

#1. Retrieve all successful bookings:

```
Create View Successful_Bookings As
SELECT * FROM bookings
WHERE Booking_Status = 'Success';
```

#2. Find the average ride distance for each vehicle type:

```
Create View ride_distance_for_each_vehicle As
SELECT Vehicle_Type, AVG(Ride_Distance) as
avg_distance FROM bookings GROUP BY
Vehicle_Type;
```

#3. Get the total number of cancelled rides by customers:

```
Create View cancelled_rides_by_customers As
SELECT COUNT(*) FROM bookings
WHERE Booking_Status = 'cancelled by Customer';
```

#4. List the top 5 customers who booked the highest number of rides:

```
Create View Top_5_Customers As
SELECT Customer_ID, COUNT(Booking_ID) as total_rides
FROM bookings
GROUP BY Customer_ID
ORDER BY total_rides DESC LIMIT 5;
```

#5. Get the number of rides cancelled by drivers due to personal and car-related issues:

```
Create View Rides_cancelled_by_Drivers_P_C_Issues As
SELECT COUNT(*) FROM bookings
WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';
```

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

```
Create View Max_Min_Driver_Rating As
SELECT MAX(Driver_Ratings) as max_rating,
MIN(Driver_Ratings) as min_rating
FROM bookings WHERE Vehicle_Type = 'Prime Sedan';
```

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#7. Retrieve all rides where payment was made using UPI:

```
Create View UPI_Payment As
SELECT * FROM bookings
WHERE Payment_Method = 'UPI';
```

#8. Find the average customer rating per vehicle type:

```
Create View AVG_Cust_Rating As
SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating
FROM bookings
GROUP BY Vehicle_Type;
```

#9. Calculate the total booking value of rides completed successfully:

```
Create View total_successful_ride_value As
SELECT SUM(Booking_Value) as total_successful_ride_value
FROM bookings
WHERE Booking_Status = 'Success';
```

#10. List all incomplete rides along with the reason:

```
Create View Incomplete_Rides_Reason As
SELECT Booking_ID, Incomplete_Rides_Reason
FROM bookings
WHERE Incomplete_Rides = 'Yes';
```

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Retrieve All Answers

#1. Retrieve all successful bookings:
Select * From Successful_Bookings;

#2. Find the average ride distance for each vehicle type:
Select * from ride_distance_for_each_vehicle;

#3. Get the total number of cancelled rides by customers:
Select * from cancelled_rides_by_customers;

#4. List the top 5 customers who booked the highest number of rides:
Select * from Top_5_Customers;
#5. Get the number of rides cancelled by drivers due to personal and car-related issues:
Select * from Rides_cancelled_by_Drivers_P_C_Issues;

#6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
Select * from Max_Min_Driver_Rating;

#7. Retrieve all rides where payment was made using UPI:
Select * from UPI_Payment;

#8. Find the average customer rating per vehicle type:
Select * from AVG_Cust_Rating;

#9. Calculate the total booking value of rides completed successfully:
Select * from total_successful_ride_value;

#10. List all incomplete rides along with the reason:
Select * from Incomplete_Rides_Reason;

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Acknowledgment / Thanking Note

I would like to extend my heartfelt gratitude to everyone who supported and guided me throughout this project. Special thanks to my mentors, peers, and anyone who provided valuable feedback and insights during the development of the OLA Data Analyst Project.

This experience has not only enhanced my technical skills in SQL and Power BI but also strengthened my ability to draw meaningful conclusions from complex datasets. I am truly grateful for the opportunity to work on this project and look forward to applying these learnings to future challenges in the field of data analytics.

— Sankalp Singh